

Limits



Direct substitution

- 1 Find $\lim_{x \rightarrow 3} (2x^2 - x + 1)$.
 - 2 Find $\lim_{x \rightarrow -1} (x^3 + 4x)$.
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Indeterminate forms: factor and cancel

- 3 Find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.
 - 4 Find $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$.
 - 5 Explain why you cannot find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.
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Limits at infinity for rational functions

- 6 Find $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.
 - 7 Find $\lim_{x \rightarrow \infty} \frac{4x + 1}{2x^2 + 3}$.
 - 8 Explain what happens to $\lim_{x \rightarrow \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.
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Multiple choice — easy

- 9 Find $\lim_{x \rightarrow -7} (3x^2)$.
 - a. 145
 - b. 147
 - c. -21
 - d. 149

10 Find $\lim_{x \rightarrow 7} (4x^2 + x - 7)$.

a. 194

b. 28

c. 198

d. 196

11 Find $\lim_{x \rightarrow -6} (2x^2 + 5x - 9)$.

a. 35

b. 31

c. -51

d. 33

12 Find $\lim_{x \rightarrow -2} (-4x^2 + 8x - 3)$.

a. -35

b. -33

c. -11

d. -37

13 Find $\lim_{x \rightarrow 2} (-3x^2 + 7x - 7)$.

a. -3

b. -5

c. -7

d. 1

14 Find $\lim_{x \rightarrow -7} (4x^2 - 2x - 9)$.

a. 203

b. 201

c. 199

d. -23

15 Find $\lim_{x \rightarrow 5} (-3x^2 - 9x + 7)$.

a. -53

b. -113

c. -115

d. -111

16 Find $\lim_{x \rightarrow 5} (4x^2 + 4x - 4)$.

a. 116

b. 114

c. 118

d. 36

17 Find $\lim_{x \rightarrow -4} (3x^2 + x + 4)$.

a. -12

b. 50

c. 46

d. 48

18 Find $\lim_{x \rightarrow -8} (3x^2 + 5x - 3)$.

a. 147

b. -67

c. 149

d. 151

19 Find $\lim_{x \rightarrow 4} (3x^2 - 9x - 1)$.

a. -25

b. 11

c. 13

d. 9

20 Find $\lim_{x \rightarrow 6} (2x^2 + 9x - 6)$.

a. 122

b. 120

c. 60

d. 118

21 Find $\lim_{x \rightarrow -5} (-4x^2 + 2x - 2)$.

a. -112

b. -110

c. 8

d. -114

22 Find $\lim_{x \rightarrow -4} (-2x^2 - 4x + 4)$.

- a. -12
- b. -10
- c. -14
- d. 28

23 Find $\lim_{x \rightarrow -3} (3x^2 - 8x - 2)$.

- a. 13
- b. 51
- c. 49
- d. 47

Multiple choice — medium

24 Find $\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{x - 3}$.

- a. 1
- b. 3
- c. -1
- d. 4

25 Find $\lim_{x \rightarrow 8} \frac{x^2 - 17x + 72}{x - 8}$.

- a. -3
- b. 1
- c. 2
- d. -1

26 Find $\lim_{x \rightarrow -5} \frac{x^2 - 4x - 45}{x + 5}$.

- a. 14
- b. -16
- c. -12
- d. -14

27 Find $\lim_{x \rightarrow 6} \frac{x^2 - 36}{x - 6}$.

- a. -12
- b. 10
- c. 12
- d. 14

28 Find $\lim_{x \rightarrow 4} \frac{x^2 - 9x + 20}{x - 4}$.

- a. -1
- b. 2
- c. -3
- d. 1

29 Find $\lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{x + 2}$.

- a. 2
- b. -1
- c. -3
- d. 1

30 Find $\lim_{x \rightarrow -2} \frac{x^2 + 2x}{x + 2}$.

a. -2

b. 2

c. -4

d. 0

31 Find $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3}$.

a. 7

b. 5

c. 3

d. -5

32 Find $\lim_{x \rightarrow -1} \frac{x^2 - 4x - 5}{x + 1}$.

a. -6

b. -8

c. 6

d. -4

33 Find $\lim_{x \rightarrow 5} \frac{x^2 - 5x}{x - 5}$.

a. -5

b. 5

c. 3

d. 7

34 Find $\lim_{x \rightarrow 6} \frac{x^2 - 2x - 24}{x - 6}$.

a. 8

b. -10

c. 12

d. 10

35 Find $\lim_{x \rightarrow 7} \frac{x^2 - 16x + 63}{x - 7}$.

a. -2

b. -4

c. 0

d. 2

36 Find $\lim_{x \rightarrow -6} \frac{x^2 + 2x - 24}{x + 6}$.

a. -10

b. -8

c. -12

d. 10

37 Find $\lim_{x \rightarrow -5} \frac{x^2 + 9x + 20}{x + 5}$.

a. -3

b. -1

c. 2

d. 1

- 38 Find $\lim_{x \rightarrow -9} \frac{x^2 + 14x + 45}{x + 9}$.
- a. -2
 - b. 4
 - c. -4
 - d. -6
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Multiple choice — hard

- 39 Find $\lim_{x \rightarrow \infty} \frac{-x^2 + 3x}{3x^2 + x}$.
- a. $\frac{1}{3}$
 - b. $\frac{2}{3}$
 - c. $-\frac{4}{3}$
 - d. $-\frac{1}{3}$
- 40 Find $\lim_{x \rightarrow \infty} \frac{-6x^2 + 6x}{8x^2}$.
- a. $-\frac{7}{4}$
 - b. $\frac{3}{4}$
 - c. $-\frac{3}{4}$
 - d. $\frac{1}{4}$
- 41 Find $\lim_{x \rightarrow \infty} \frac{-7x^2 + 3x}{-5x^2 + 4x}$.
- a. $\frac{7}{5}$
 - b. $-\frac{7}{5}$
 - c. $\frac{2}{5}$
 - d. $\frac{12}{5}$

42 Find $\lim_{x \rightarrow \infty} \frac{-6x^2 + x}{-4x^2 - 4x}$.

a. $\frac{3}{2}$

b. $-\frac{3}{2}$

c. $\frac{5}{2}$

d. $\frac{1}{2}$

43 Find $\lim_{x \rightarrow \infty} \frac{-8x^2 + 2x}{9x^2 + x}$.

a. $-\frac{17}{9}$

b. $\frac{1}{9}$

c. $-\frac{8}{9}$

d. $\frac{8}{9}$

44 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+81} - \sqrt{81}}{x}$.

a. $\frac{1}{18}$

b. $\frac{1}{9}$

c. $\frac{1}{6}$

d. $\frac{2}{9}$

45 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+64} - \sqrt{64}}{x}$.

a. $\frac{1}{4}$

b. $\frac{1}{8}$

c. $\frac{1}{16}$

d. $\frac{3}{16}$

46 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+144} - \sqrt{144}}{x}$.

a. $\frac{1}{6}$

b. $\frac{1}{8}$

c. $\frac{1}{24}$

d. $\frac{1}{12}$

47 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+25} - \sqrt{25}}{x}$.

a. $\frac{2}{5}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$

d. $\frac{1}{5}$

48 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+49} - \sqrt{49}}{x}$.

a. $\frac{2}{7}$

b. $\frac{3}{14}$

c. $\frac{1}{14}$

d. $\frac{1}{7}$

49 Find $\lim_{x \rightarrow \infty} \frac{x^3 + 5x}{9x^2 - 9x}$ (state whether it is ∞ or $-\infty$).

a. $\frac{1}{9}$

b. 0

c. $-\infty$

d. ∞

50 Find $\lim_{x \rightarrow \infty} \frac{-9x^3 - 8x}{-8x^2 + 2x}$ (state whether it is ∞ or $-\infty$).

a. ∞

b. $\frac{9}{8}$

c. $-\infty$

d. 0

51 Find $\lim_{x \rightarrow \infty} \frac{-2x^3 - x}{3x^2 + 9x}$ (state whether it is ∞ or $-\infty$).

a. $-\frac{2}{3}$

b. 0

c. ∞

d. $-\infty$

52 Find $\lim_{x \rightarrow \infty} \frac{6x^3 + 7x}{2x^2 + 6x}$ (state whether it is ∞ or $-\infty$).

a. 0

b. 3

c. ∞

d. $-\infty$

53 Find $\lim_{x \rightarrow \infty} \frac{2x^3 - 8x}{9x^2 - 3x}$ (state whether it is ∞ or $-\infty$).

a. 0

b. $\frac{2}{9}$

c. $-\infty$

d. ∞